


```

ble
em:          7940748      517752      6777396      3564      800208      742
996
wap:         2097152      0      2097152
eja@LAPTOP-FGDPB110:~$ ps
  PID TTY          TIME CMD
   463 pts/0    00:00:00 bash
   477 pts/0    00:00:00 ps
eja@LAPTOP-FGDPB110:~$ top
top - 06:01:23 up 7 min,  1 user,  load average: 0.09, 0.04, 0.01
tasks: 22 total,  1 running, 21 sleeping,  0 stopped,  0 zombie
Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7754.6 total,  6617.9 free,   506.2 used,   781.5 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,    0.0 used.  7248.4 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+
    1 root        20   0   21848  13128  9796  S   0.0   0.2   0:01.68
    2 root        20   0    3180   2208  2072  S   0.0   0.0   0:00.03
    6 root        20   0    3196   2124  2008  S   0.0   0.0   0:00.01
   61 root        19  -1  42172  16200 15128  S   0.0   0.2   0:00.30
  112 root        20   0   25020   6408  5148  S   0.0   0.1   0:00.38
  126 systemd+    20   0   21344  13232 11108  S   0.0   0.2   0:00.24
  127 systemd+    20   0   91036   7984  7024  S   0.0   0.1   0:00.16
  190 root        20   0    4244   2688  2432  S   0.0   0.0   0:00.02
  191 message+    20   0    9652   5364  4660  S   0.0   0.1   0:00.12
  198 root        20   0   17976   8820  7780  S   0.0   0.1   0:00.14
  200 root        20   0 1756628  13716 11444  S   0.0   0.2   0:00.18
  208 syslog      20   0   222516  5672  4624  S   0.0   0.1   0:00.12
  216 root        20   0    3124   1972  1836  S   0.0   0.0   0:00.01
  222 root        20   0  107036  23092 13636  S   0.0   0.3   0:00.31
  330 root        20   0    6700   4664  3884  S   0.0   0.1   0:00.02
  377 teja        20   0   20324  11516  9392  S   0.0   0.1   0:00.25
  380 teja        20   0   21164   3572  1848  S   0.0   0.0   0:00.00
  404 teja        20   0    6080   5364  3696  S   0.0   0.1   0:00.05
  459 root        20   0    3184   1120    980  S   0.0   0.0   0:00.00
  461 root        20   0    3200   1256  1108  S   0.0   0.0   0:00.00
  463 teja        20   0    6080   5368  3648  S   0.0   0.1   0:00.12

```

```
eja@LAPTOP-FGDPB110:~$ history
1  cffftcvgvgvvvvvvvvvv
2  clear
3  sudo apt update
4  sudo apt install build-essential installing:
5  gcc --version
6  gcc process . c -o process
7  ./process
8  clear
9  mkdir OSSP
10 cd OSSP
11 git init
12 mkdir src include obj bin docs tests scripts assets
13 touch .gitignore README.md LICENSE Makefile
14 touch src/main.c
15 touch src/shell.c
16 touch src/parser.c
17 touch src/executor.c
18 touch src/builtin.c
19 touch include/shell.h
20 touch include/parser.h
21 touch include/executor.h
22 touch include/builtin.h
23 touch docs/design.md
24 touch docs/report.pdf
25 touch tests/test_parser.c
26 touch tests/test_executor.c
27 touch scripts/run.sh
28 touch assets/architecture.png
29 tree
30 sudo apt install tree
31 tree
32 sudo apt install tree
33 sudo apt install build-essential
34 gcc --version
35 nano process.c
36 sudo apt install tree
```

```
48 cal
49 hostname
50 uname -a
51 exit
52 sudo apt update
53 sudo apt upgrade
54 sudo apt install gcc
55 gcc --version
56 nano process.c
57 gcc process.c -o process
58 ./process
59 ls
60 pwd
61 date
62 cal
63 whoami
64 hostname
65 uname
66 uname -a
67 ps
68 top
69 history
70 Ctrl + O
71 mkdir Week3
72 cd Week3
73 nano task1.c
74 Ctrl + O~
75 hello
76 mkdir OSSP
77 cd OSSP
78 mkdir src include obj bin docs tests scripts assets
79 mkdir OSSP
80 cd OSSP
81 mkdir src include obj bin docs tests scripts assets
82 ls
83 ls -l
84 find . -maxdepth 3
85 touch .gitignore README.md LICENSE Makefile
86 touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c
87 touch include/shell.h include/parser.h include/executor.h include/builtin.h
88 touch docs/design.md docs/report.pdf
```

```
94 git status
95 git add .
96 git status
97 git commit -m "Initial OSSP project structure"
98 git status
99 nothing to commit, working tree clean
100 nano fork_example.c
101 gcc fork_example.c -o fork_example
102 ./fork_example
103 ls
104 pwd
105 data
106 whoami
107 uname
108 uname -r
109 id
110 tty
111 uptime
112 users
113 who
114 w
115 groups
116 env
117 printenv
118 df
119 free
120 ps
121 top
122 df
123 free
124 ps
125
```

```
108 uname -r
109 id
110 tty
111 uptime
112 users
113 who
114 w
115 groups
116 env
117 printenv
118 df
119 free
120 ps
121 top
122 df
123 free
124 ps
125 top
126 free
127 ps
128 top
129 vmstatvmsat
130 vmstat
131 iostat
132 lscpu
133 history
```

[illegible]

```

bhavan@LAPTOP-FGDPB110:~$ sudo apt install sysstat
bhavan@LAPTOP-FGDPB110:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:         0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                  6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4992.02
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pa
pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp l
constant_tsc rep_good nopl xtopology tsc_known_freq nonstop_tsc cpui
tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2
x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand
hypervisor_lahf_lm abm 3dnowprefetch cpuid_fault epb ibrs ibpb stib
ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx
smep bmi2 erms invpcid rdt_a rdseed adx smap clflushopt clwb sha_n
xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vae
vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt
flush_lld arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                           7.5 MiB (6 instances)
L3:                           12 MiB (1 instance)
NUMA:
NUMA node(s):                 1
NUMA node0 CPU(s):            0-11
Vulnerabilities:
Gather data sampling:         Not affected
Ghostwrite:                   Not affected
Indirect target selection:     Not affected
bhavan@LAPTOP-FGDPB110:~$

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127  ps
128  top
129  vmstat
130  vnstat
131  iostat
132  lscpu
133  history
bhavan@LAPTOP-FGDPB110:~$ clear

```

```
bhavan@LAPTOP-FGDPB110:~$ echo hello
hello
bhavan@LAPTOP-FGDPB110:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundati
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
factor
0
factor 12
(standard_in) 2: syntax error
yes
0
sleep
0
```

echo command

The echo command is used to **display a line of text or a value on the terminal**. In the practical, echo hello displays hello on the screen.

2. bc command

bc stands for **Basic Calculator**. It is a command-line calculator used to perform **arithmetic and mathematical calculations** in Linux. When bc is executed, it opens an interactive calculator.

3. factor command in bc

The factor operation is used to find the **prime factors of a given integer**. It can be used for mathematical calculations inside the bc environment.

4. yes command

The yes command repeatedly outputs the given text until it is stopped. If no text is specified, it continuously displays y.

5. sleep command

The sleep command is used to **pause or delay execution for a specified amount of time**. For example, sleep 5 pauses execution for 5 seconds